Project Topic – Identifying water sources in Canada (rivers, lakes, ponds) for interactive map creation

Project Plan – Phase 1 (Backend):

Objectives: pick the most valuable dataset for water identification according to rules listed below. Identify limitations, what to use as training data (varied environments), how to identify water compared to other image features (simple compared to other project topics), identify what tools can be used as well as what they can be used for.

Water identification (rules): water reflects some blue/green light but strongly absorbs near-infrared (NIR, Sentinel-2 band B8) and shortwave-infrared (SWIR, bands B11/B12), so it appears dark in those bands while land and vegetation appear bright. Use NDWI = (B3 − B8) / (B3 + B8), where values above ~0 indicate water. Classify each pixel as water or non-water and use shape and context to reject look-alikes such as cloud or terrain shadows, wet soil, asphalt, and ice. Use summer imagery (June–September) to avoid frozen water and consistent results.

Feature identification (different water sources): Rivers are narrow and long, while lakes are deep(potentially), with ponds being smaller bodies of water and lakes being larger bodies of water, like ponds in shape and style. Ponds more often have vegetation growing on top of water than lakes. You can image looking for a thick line (river) vs round area (pond/lake).

Limitations: Any water sources that are smaller than the resolution size of the dataset (10m most likely). Water that is covered by large tree canopies. Super shallow water (potentially?). Smaller ponds have the chance of drying out in the summer depending on weather conditions and pond depth. Persistent cloud coverage weakens the usability of some areas.

Possible datasets:

Dataset	Resolution	Canada Coverage	Where	Notes

Sentinel-2 SR	***10m***	***Full***	***GEE COPERNICUS/S2_SR_HARMONIZED***	***NIR/SWIR bands for easy water separation***
Landsat 8/9	30m	Full, all the way to 1980s	GEE	Coarser, decades of informatio n
Sentinel-1 SAR	10m	Full	GEE COPERNICUS/S1_GRD	Sees through clouds, water appears dark, confused by wind-roughened water and ice
Planetscop e	3m	Full	Planet Education program	Licensing delay, free for students

Possible pre-labelled sets to use for AI training:

Dataset	Resolution	Where	Notes
Dynamic World	10m	GEE GOOGLE/DYNAMICWORLD/V1	Per-scene water class matching Sentinel-2 dates (main training set)
ESA WorldCover	10m	GEE ESA/WorldCover/v200	Single 2021 snapshot
JRC Global Surface Water	30m	GEE JRC/GSW1_4/GlobalSurfaceWater	Water occurrence since 1984 (to flag ponds that dry up)
National Hydro Network	Vector	open.canada.ca	Official Canadian lakes and rivers (to

			check classification)

Possible Ai models:

Model	Type	Strengths	Weaknesses
U-Net	***Segmentation***	***Standard for satellite water mapping, trains on colab***	***Less context over large areas than newer models***
DeepLabV3+	***Segmentation***	***Good comparison model***	***Slower and heavier than U-Net***
SegFormer	Transformer segmentation	Strong accuracy, captures wider context	Needs more data and tuning
YOLOv8/v11	Detection and instance segmentation	Fast, outputs boxes and masks	Built for distinct objects, weak on smaller features
Prithvi	Satellite foundation model	Pretrained on satellite imagery, has flood and water fine tuning	Heavy, uses specific HLS input bands, harder setup
SAM/SAM 2	General segmentation	Can outline objects from a click or box with no training	Not trained for spectral bands, needs prompts
Random Forest on NDWI/bands	Classic machine learning	Runs inside GEE, very fast	Pixel by pixel, no shape context
NDWI threshold	***Rule-based***	***No training***	***Fails on shadows, turbid water***

Training data: use images from specific locations in Canada that features varied landscapes such as large rivers (more rural), tundras (Nunavut maybe), urban areas with larger water sources, great lakes and shield lakes. Does not need to feature thousands of locations across Canada, just the ones that show up the most and the areas that are most common.

- GEE (export images/tile masks for water to use as the training set)

- Labelme (trustworthy small testset pulled from the GEE dataset from specific regions for testing the AI model)

Project flow:

1. Get exported data from GEE and Labelme tiles as training set and test set
2. Split data into training, validation and test sets
3. Load data into Pytorch for training with chosen AI model/train model
4. Evaluate trained Ai and possibly repeat training
5. Take data given from AI model and plug into scikit-image (Clean up this data (noise), group pixels into objects that can be classified, take rough measurements for classification, classify based on rules (area and compactness)).

Next steps:

- Get training data ready to use (write GEE python code to export specified regions)
- Determine storage location (google drive cloud, locally)
- Train AI model
- Add to the Github repo